

Kyle Medina

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EDUCATION

University of California, Los Angeles - *B.S., Aerospace Engineering* | June 2027 | GPA: 3.31 **Los Angeles, CA**
• **Relevant Coursework:** Rocket Propulsion Systems; Aerodynamics; Intermediate Fluid Dynamics; Thermodynamics; Transport Phenomena; Dynamic Systems Control; Modeling & Analysis of Dynamic Systems; Space Technology; Dynamics of Particles & Rigid Bodies; Statics & Strengths of Materials; Science of Engineering Materials; Electrical & Electronic Circuits
In Progress: Aircraft Propulsion Systems; Analysis of Aerospace Structures; Flight Mechanics, Stability, & Control of Aircraft
Lund University - *Exchange Semester, Aerospace Engineering* | August 2025 - December 2025 **Lund, Sweden**

ENGINEERING PROJECTS & EXPERIENCE

Rocket Project at UCLA - *Engine Nozzle Lead & Test Engineer* | Sept 2024 - June 2026 **Los Angeles, CA**
• Led nozzle design trade studies and CAD modeling; coordinated with injector, chamber, and ignitor subsystem leads to ensure interface compatibility and integration readiness
• Researched, designed, and manufactured a static fire test stand; fabricated custom components using lathe, drill press, and mill
• Conducted injector water flow testing to characterize flow uniformity and pressure drop; performed hydrostatic testing on engine chambers to verify structural integrity
Airplane Engineering Design - *Team Leader, 4-Person Team* | April 2024 - June 2024 **Los Angeles, CA**
• Led full-cycle design and build of an RC aircraft, using CAD, MATLAB, and XFLR5 for aerodynamic modeling and control
• Completed a successful self-propelled takeoff, 2.5-minute flight with two barrel rolls, 80+ ft/s top speed, and safe landing
Independent Projects **San Diego, CA**
Motorcycle Build - Refurbished Bicycles - Designed & Built Custom Furniture - Rebuilt Suspension & Brakes on Vehicles

RESEARCH EXPERIENCE

Thermal Science Laboratory for Semiconductor Packaging & Integration, UCLA **Los Angeles, CA**
Undergraduate Research Assistant | April 2026 - Present
• Lead development of a scalable, batch-capable process to bond PTFE/PP membranes to 3D-printed resin manifolds for a two-phase electronics cooling device, replacing a slow manual adhesive process that left gaps
• Wrote bonding requirements from operating conditions (110 °C hot-wet service, leak-tight at 72 kPa, adhesive-free active area) and derived process pressure limits from material datasheets
• Ran a trade study of four adhesive families and application methods (stencil, die-cut), down-selected two candidates for bench testing beginning fall 2026, and defined a verification plan with dye-penetrant leak testing and 70-100 °C thermal cycling
Cai Group & TEMP Lab at UC San Diego - *Undergraduate Research Assistant* | **Summers 2025 & 2026** **La Jolla, CA**
• Characterized membrane permeability across 40 flow trials; built a MATLAB analysis pipeline with Monte Carlo uncertainty propagation and automated data-validity checks
• Root-caused anomalous test results to test-rig flow resistance rather than sample properties; recommended a blank-run calibration and randomized test order to correct the method
• Developed the lab's repeatable process for fabricating uniform ~25 µm hydrogel coatings on porous glass fiber membranes, plus a microscopy-based method to validate coating thickness

LEADERSHIP & ACTIVITIES

Engineering Ambassadors - *Ambassador* | April 2024 - Present **Los Angeles, CA**
• Represent UCLA Engineering at 25+ tours, 1-on-1 Q&As, 6 Commencement Ceremonies, and 2 Welcome Days
• Deliver engineering presentations at Santa Monica College; collaborate with faculty and donors to promote STEM engagement

SKILLS

• **Software & Tools:** Python, Matlab, CAD/SolidWorks, XFLR5, Microsoft Office, Google Workspace
• **Fabrication & Lab:** Lathe, Mill, Drill Press, Band Saw, 3D Printing (filament & resin), Laser Cutting, Microscope, Micropipette, Flow Loop & Permeability Testing
• **Competencies:** Requirements Definition, Trade Studies, Test Planning, Root Cause Analysis, Leadership, Communication

PROFESSIONAL EXPERIENCE

San Diego Rock Supply - *Operator/Trainer/Cashier* | *El Cajon, CA* | May 2023 – September 2023 **El Cajon, CA**
• Operated a CAT 906 Front Loader; Trained new hires; Oversaw Deliveries, Machinery Maintenance, & Equipment lock-up
Vons - *Courtesy Clerk* | *Lake Murray, CA* | July 2024 – September 2024 **Lake Murray, CA**
• Maintained store cleanliness and organization; Supported front-end operations; Provided excellent customer service